Blockchain dApp Development (UQ24CA343AA3)

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Unit I- Introduction to Blockchain

Blockchain: Introduction, Working, Origins, P2P Network, Blockchain Vs Cryptocurrency

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Why BlockChain?

Think About This…

Imagine you want to transfer ₹1000 to your friend:

How does it happen?

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Why BlockChain?

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Why BlockChain?

Problems with Traditional Systems:

- Dependence on Intermediaries
- Transaction Delays
- Transaction Fees
- Single Point of Failure
- Limited Transparency
- Risk of Data Tampering
- Limited Availability
- Security Risks

To overcome these problems, Blockchain was introduced

What is BlockChain?

The concept is introduced by **Satoshi Nakamoto 2009** in cryptocurrency Bitcoin

Technical definition:Blockchain is a peer-to-peer, distributed ledger that is cryptographically-secure, append-only, immutable (extremely hard to change), and updateable only via consensus or agreement among peers.

or

Blockchain is a decentralized digital ledger that securely records transactions in linked blocks using cryptographic techniques.

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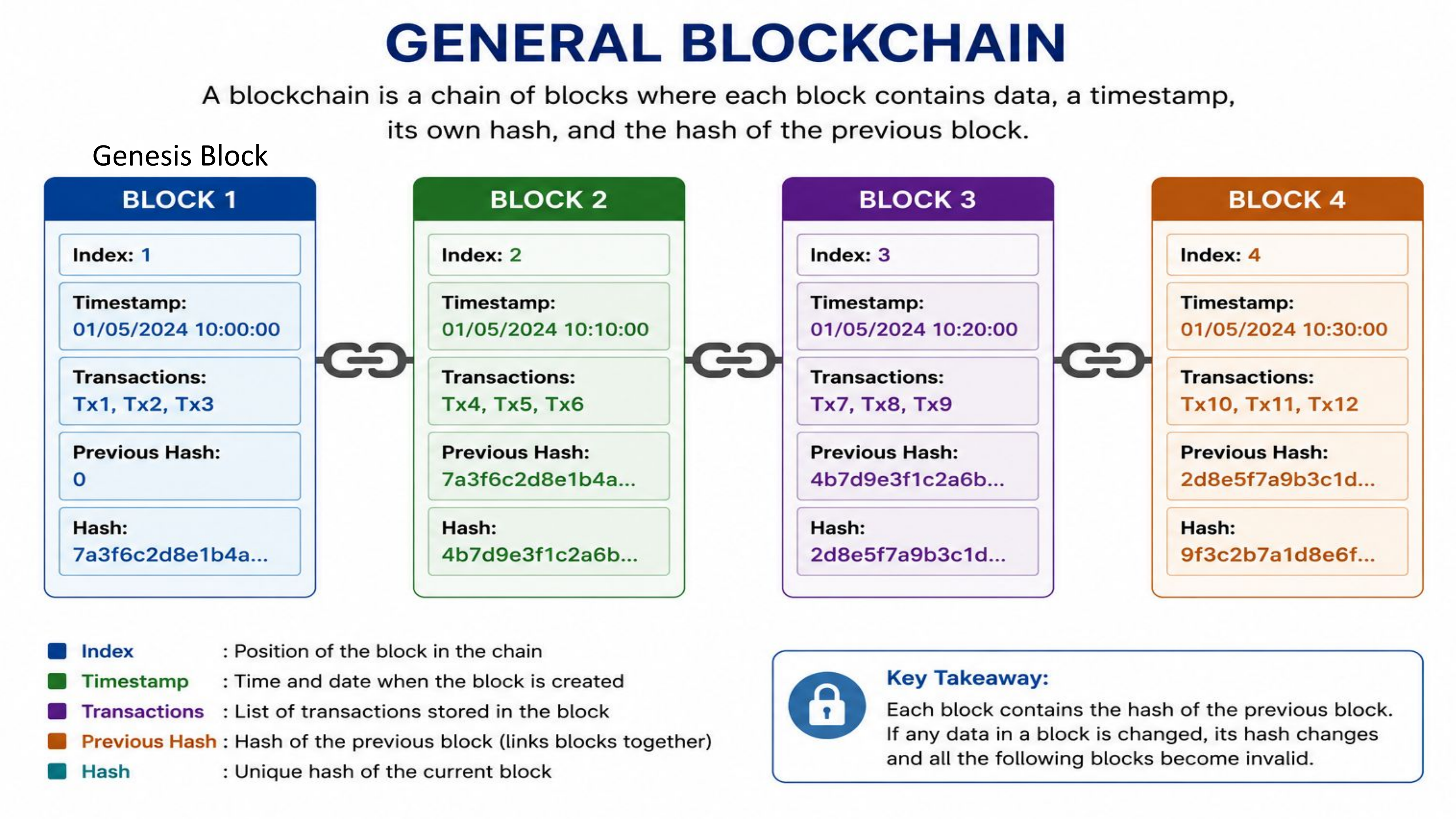
Why it is called BlockChain?



Features of BlockChain

- Decentralized
- Distributed Ledger
- Transparency
- Security
- Immutability
- Traceability

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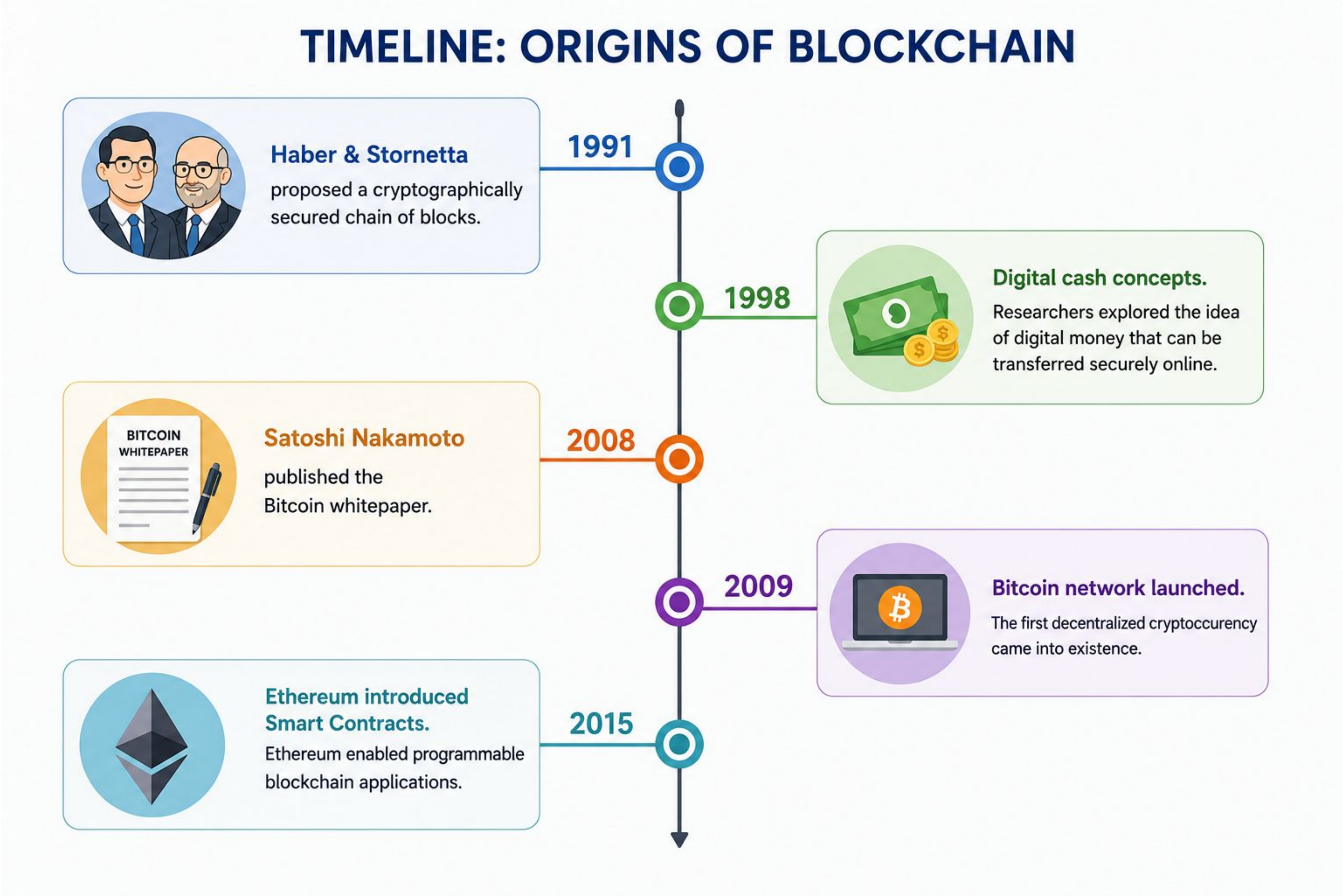
Origins of BlockChain



Why Was Blockchain Invented?

The main objective was

- Remove intermediaries
- Build trust without banks
- Secure digital transactions
- Prevent double spending

CAP theorem:

- Consistency
- Availability
- Partition tolerance

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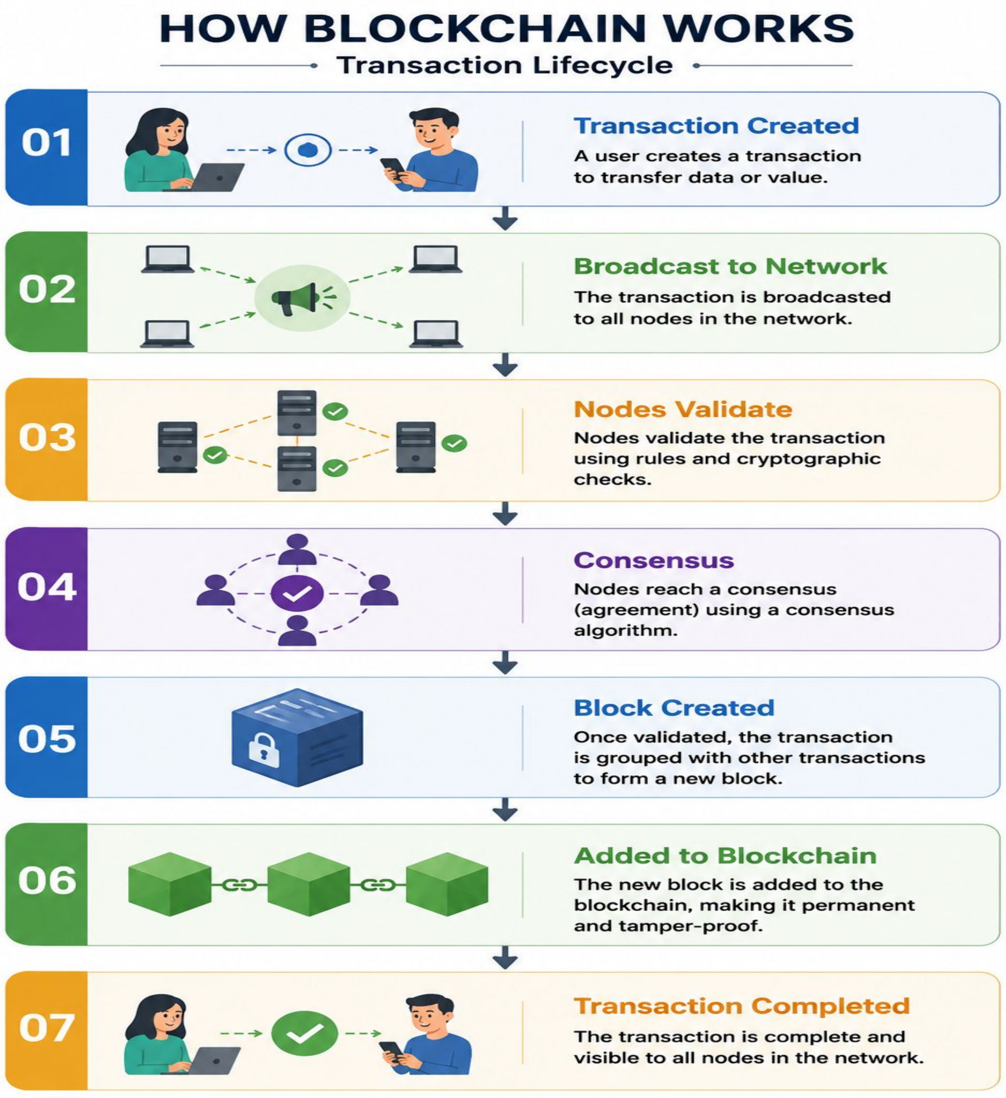
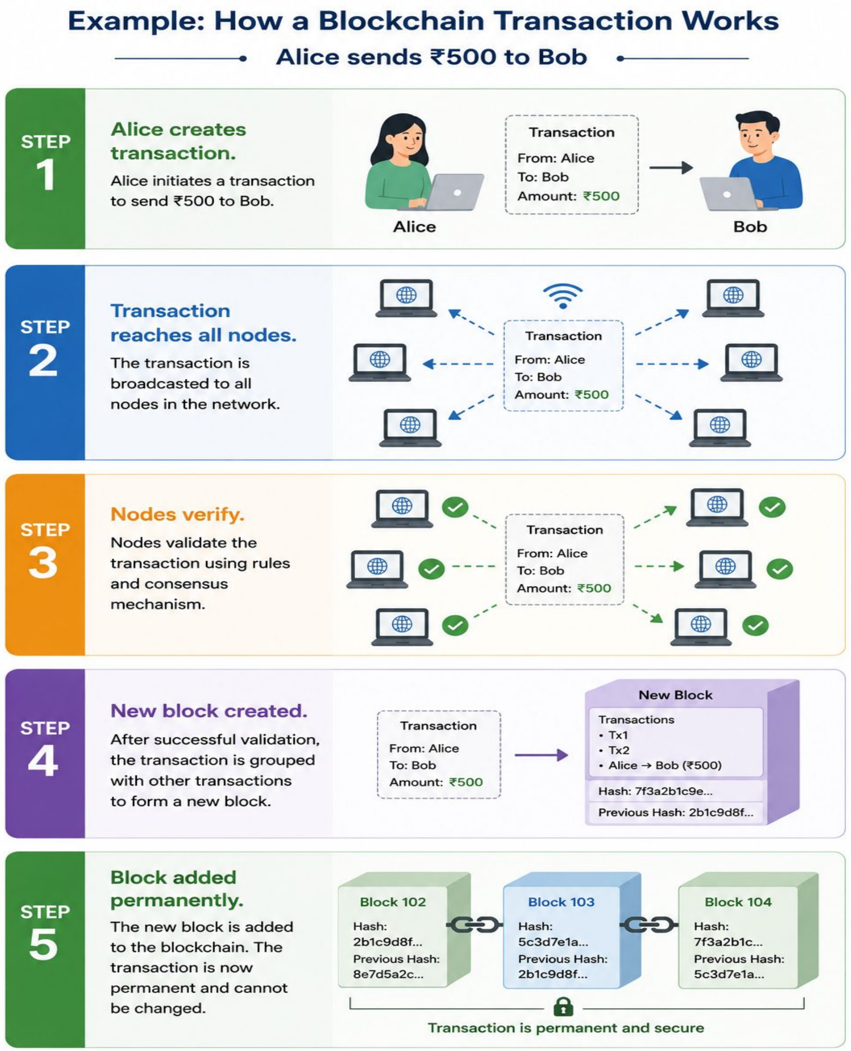
Working of Blockchain





Types of Network/Systems

- Centralized systems
- Distributed system
- Decentralized system

Centralized systems

- Managed by a **single central authority or server**
- Users depend on that central entity for communication and data management

Characteristics:

- One central authority controls the system.
- Easy to manage and monitor.
- Single point of failure.
- Lower transparency.

Examples:

- Banks
- Facebook
- Gmail

Decentralized systems

- **Multiple independent authorities or servers** that share control and responsibilities

Characteristics:

- Control is shared among multiple entities.
- Better reliability than centralized systems.
- Reduced dependency on a single authority.
- Improved fault tolerance.

Examples:

- Multiple bank branches
- Regional government offices
- Consortium blockchain

Distributed systems

- Network of **interconnected computers (nodes)** that work together.
- Node stores data and performs tasks without relying on a central server

Characteristics:

- No central authority.
- Every node has a copy of the data.
- High availability and scalability.
- Fault tolerant and highly secure.

Examples:

- Bitcoin
- Ethereum
- Blockchain networks
- BitTorrent

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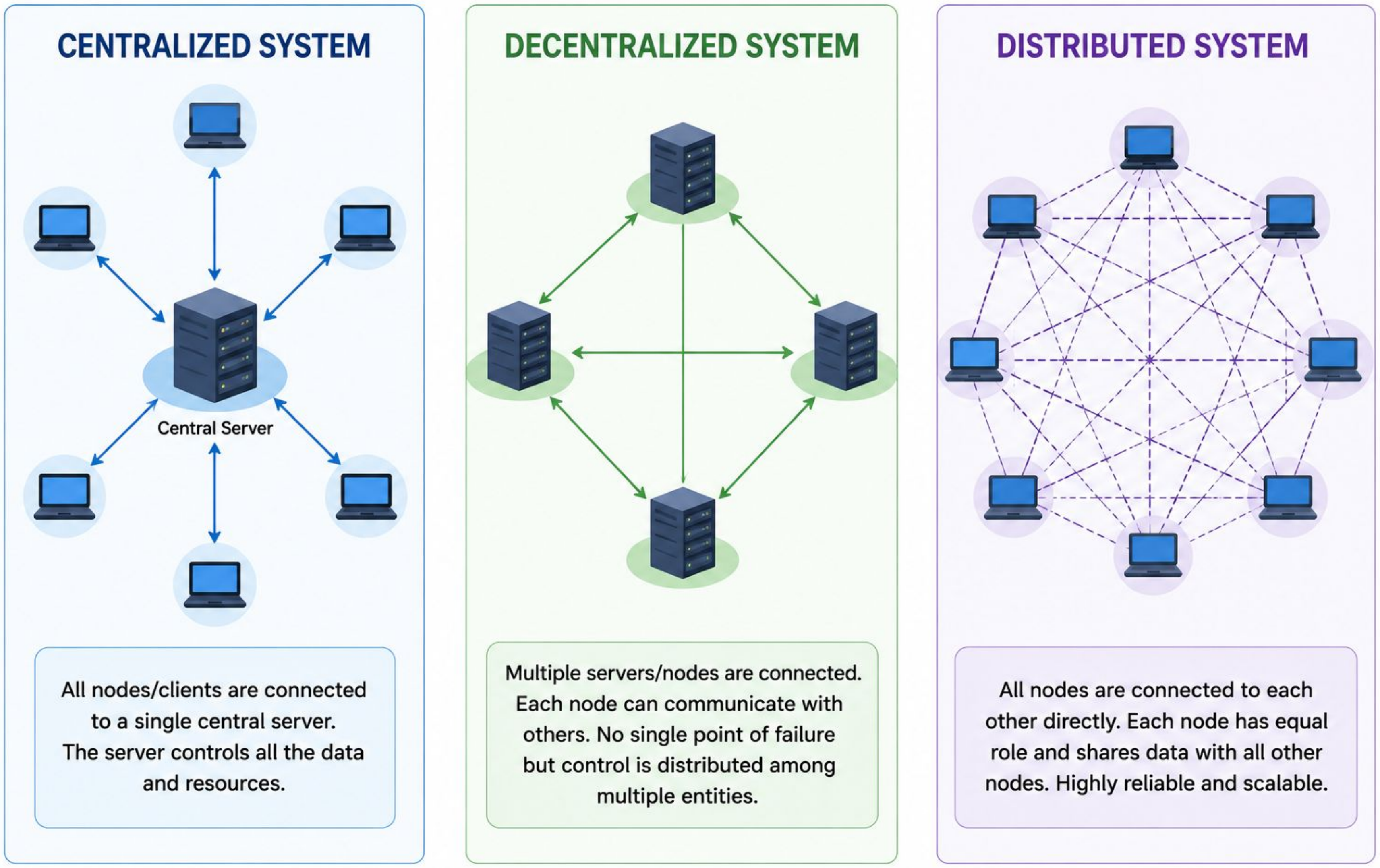
Types of Network/Systems



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Comparison of different systems

Feature	Centralized	Decentralized	Distributed
Control	Single Authority	Multiple Authorities	No Central Authority
Decision Making	Central Server	Shared Among Organizations	Shared by All Nodes
Single Point of Failure	Yes	Reduced	No
Data Storage	One Central Database	Multiple Databases	Data Replicated Across All Nodes
Transparency	Low	Moderate	High
Examples	Bank, Facebook	Consortium Blockchain	Bitcoin, Ethereum

Distributed Peer-to-Peer (P2P) Network

A **Distributed Peer-to-Peer (P2P) network** is a network in which **multiple computers (nodes)** are connected directly and work together without relying on a central server.

Advantages of a Distributed P2P Network

- ✔ Eliminates a single point of failure.
- ✔ Improves security and data integrity.
- ✔ Increases transparency and trust.
- ✔ Provides high availability and reliability.
- ✔ Makes the network resistant to tampering and cyberattacks.

Why is Blockchain Distributed?

- No Central Authority
- Shared Ledger
- Peer-to-Peer Communication
- Consensus-Based Validation
- Fault Tolerance

Blockchain vs Cryptocurrency

Blockchain	Cryptocurrency
A decentralized digital ledger technology	A digital or virtual currency
Stores and secures transaction records	Used for digital payments and value transfer
Acts as the underlying technology/platform	An application built on blockchain technology.
Supports multiple applications beyond finance.	Primarily used for financial transactions
Can exist without cryptocurrency	Cannot exist without blockchain.
Focuses on transparency, security, and decentralization	Focuses on digital asset exchange.
Examples: Ethereum, Hyperledger Fabric, Corda	Examples: Bitcoin (BTC), Ether (ETH), Litecoin (LTC)

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Key Takeaways

- The need for Blockchain technology.
- The concept and key features of Blockchain.
- The origin and evolution of Blockchain.
- How Blockchain works through transaction validation and block creation.
- Why Blockchain is a Distributed Peer-to-Peer (P2P) network.
- The difference between Blockchain and Cryptocurrency.
- The four types of Blockchain
- Major applications of Blockchain across various industries.

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Types, Applications, Evolution of BT

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Types of Blockchain

Blockchain networks are classified into four types based on who can access and manage the network.

- Public Blockchain
- Private Blockchain
- Consortium Blockchain
- Hybrid Blockchain

Public Blockchain

- Open to everyone.
- Anyone can join, read, and validate transactions.
- No central authority.
- Highly transparent and secure.
- Suitable for public applications.

Examples: Bitcoin, Ethereum

Private Blockchain

- Access is restricted to authorized users.
- Controlled by a single organization.
- Faster than public blockchain.
- Provides better privacy and security.
- Used within organizations.

Examples: Hyperledger Fabric, R3 Corda

Consortium Blockchain

- Managed by a group of organizations.
- Only selected members can participate.
- Shared control and decision-making.
- More secure and efficient than public blockchain.
- Commonly used for business collaborations.

Examples: IBM Food Trust, Marco Polo Network

Hybrid Blockchain

- Combination of public and private blockchain.
- Some data is public, while sensitive data remains private.
- Offers flexibility, privacy, and transparency.
- Organizations decide what information to share publicly.

Examples: XinFin (XDC Network), Dragonchain

Comparison of Types of Blockchain

Feature	Public	Private	Consortium	Hybrid
Access	Anyone	Authorized Users	Selected Organizations	Public + Private
Control	No Single Owner	One Organization	Multiple Organizations	Shared Control
Transparency	High	Low	Medium	Medium
Speed	Moderate	High	High	High
Examples	Bitcoin, Ethereum	Hyperledger	IBM Food Trust	XinFin, Dragonchain

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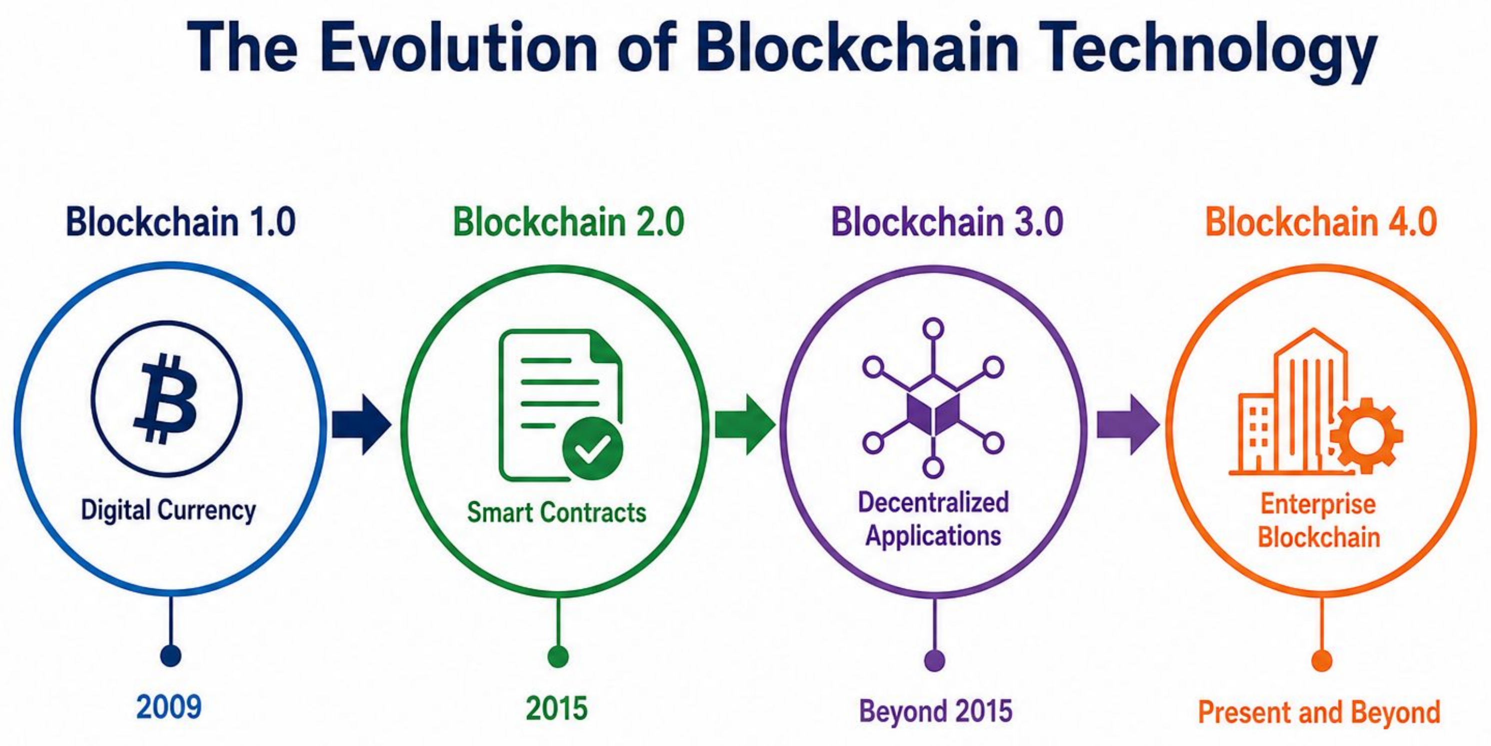
Evolution of Blockchain



Blockchain 1.0 – Digital Currency (2009)

The first generation of blockchain was introduced with **Bitcoin** by **Satoshi Nakamoto** in **2009**.

Key Features

- Focused on digital currency.
- Enabled peer-to-peer financial transactions.
- Eliminated the need for banks as intermediaries.
- Transactions are transparent and secure.

Applications

- Bitcoin
- Digital payments
- Cross-border money transfer

Blockchain 2.0 – Smart Contracts (2015)

The second generation introduced **Smart Contracts**.

Key Features

- Programmable blockchain.
- Automated contract execution.
- Supports decentralized applications (DApps).
- Reduced manual intervention.

Applications

- Decentralized Finance (DeFi)
- Insurance
- Supply Chain
- Digital Identity

Blockchain 3.0 – Decentralized Applications

Blockchain expanded beyond finance into sectors such as healthcare, education, government, and logistics.

Key Features

- Development of Decentralized Applications (DApps).
- Improved scalability and performance.
- Better user experience.
- Cross-industry adoption.

Applications

- Healthcare
- Education
- Voting Systems
- Real Estate
- Gaming

Blockchain 4.0 – Enterprise Blockchain

The latest generation focuses on integrating blockchain with modern technologies to solve real-world business problems.

Key Features

- Enterprise adoption.
- High scalability.
- Integration with AI, IoT, and Cloud Computing.
- Faster and energy-efficient consensus mechanisms.

Applications

- Smart Cities
- Industry 4.0
- Supply Chain Management
- Healthcare
- Banking

Applications of Blockchain

1. Banking and Finance
2. Supply Chain Management
3. Healthcare
4. Education
5. Voting Systems
6. Real Estate
7. Insurance
8. E-Commerce
9. Government Services
10. Internet of Things (IoT)

Key Takeaways

- The types of Blockchain
- Major applications of Blockchain across various industries.
- The evolution of Blockchain

Unit I- Introduction to Blockchain

Cryptography in Blockchain, Merkle Tree and Hashing

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Cryptography

Cryptography is the technique of securing data by encrypting it so that only authorized users can read or access it.

Purpose:

- Secure communication
- Data confidentiality
- Authentication
- Integrity

Why Blockchain Needs Cryptography?

- Protects transaction data.
- Verifies user identity.
- Prevents unauthorized access.
- Ensures data integrity.
- Makes blockchain tamper-resistant.

Goals of Cryptography

- Confidentiality
- Integrity
- Authentication
- Non-repudiation

Types of Cryptography

- Symmetric Key Cryptography
- Asymmetric Key Cryptography
- Hash Functions

Cryptography in Blockchain

- Digital Signatures
- Public & Private Keys
- Wallets
- Hashing
- Block Verification

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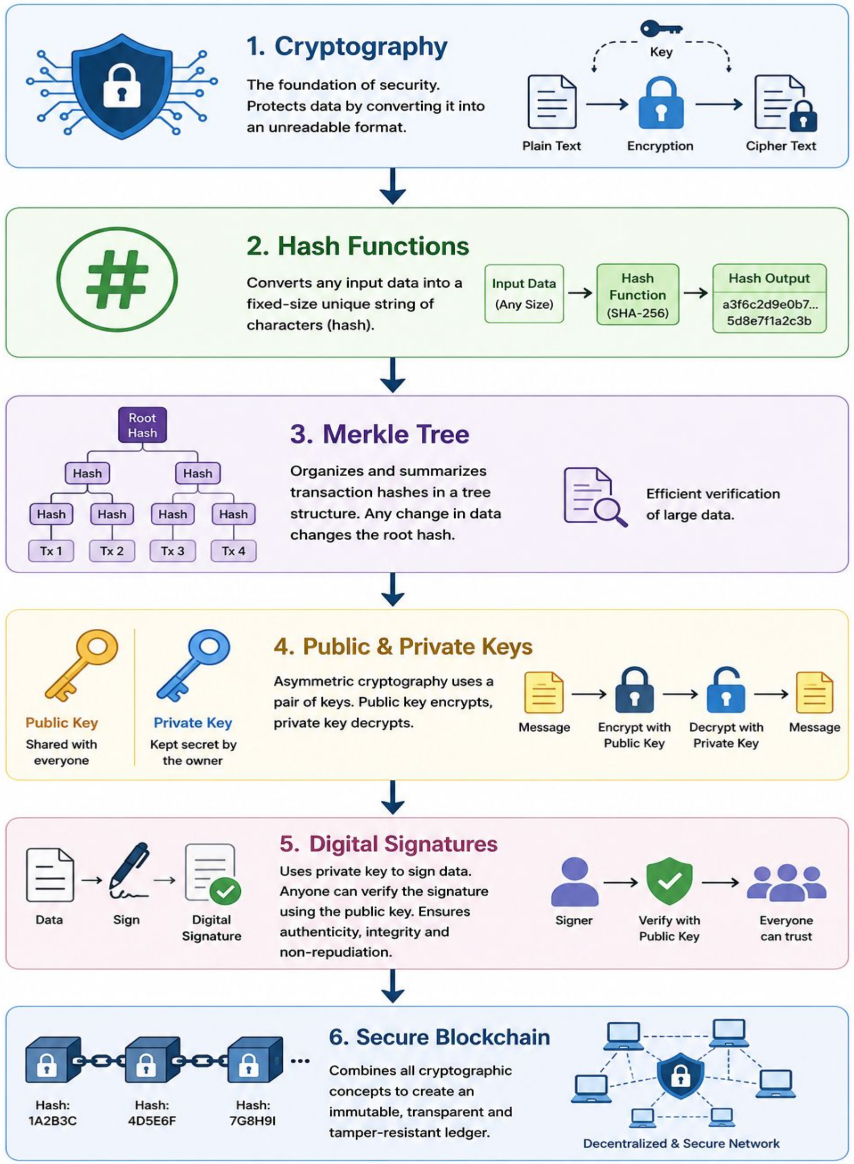
How Cryptography Secures Blockchain



Hashing

The process of converting data of any size into a fixed-length unique value called a hash using a mathematical algorithm called a hash function.

- Produces a fixed-length output (hash value).
- Same input always gives the same hash.
- A small change in input produces a completely different hash.
- One-way process (cannot be reversed).
- Used to verify data integrity.

Characteristics of Hash Functions

- Deterministic
- Fixed-Length Output
- Fast Computation
- Avalanche Effect
- One-Way Function
- Collision Resistant

Hashing in Blockchain

- Generates a unique identity for every block.
- Links blocks using the previous block's hash.
- Detects unauthorized changes to data.
- Verifies transaction integrity.
- Secures blockchain records.

Merkle Tree

Hierarchical data structure that organizes transaction hashes into a tree and combines them to produce a single hash called the Merkle Root.

- Stores hashes of all transactions.
- Reduces the amount of data needed for verification.
- Makes transaction verification faster.
- Ensures data integrity.
- Widely used in blockchain systems.

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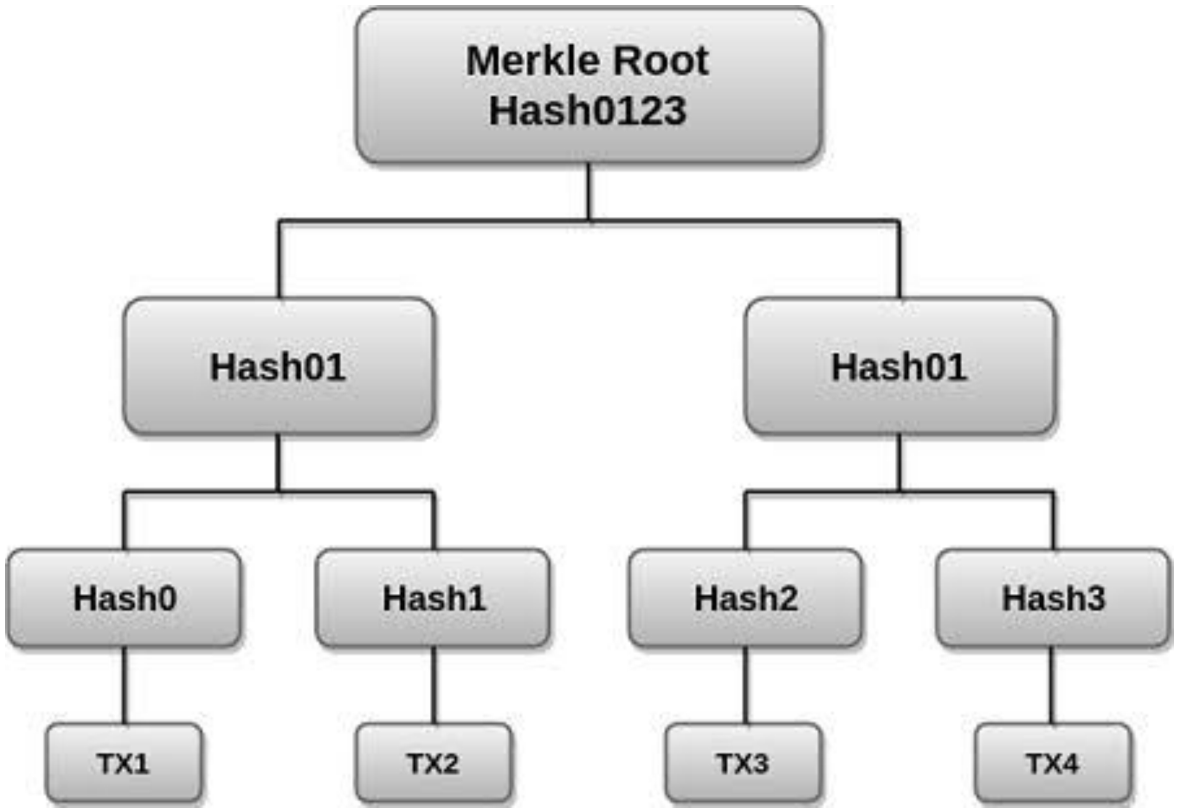
Structure of a Merkle Tree



Advantages of Merkle Tree

- Faster transaction verification.
- Improves blockchain efficiency.
- Detects data tampering.
- Reduces storage and bandwidth requirements.
- Enhances security and scalability.

Hashing vs Merkle Tree

Hashing	Merkle Tree
Converts data into a unique hash value.	Organizes multiple hashes into a tree structure.
Verifies individual data integrity.	Verifies multiple transactions efficiently.
Produces a single hash for one input.	Produces one Merkle Root for many transactions.
Used for securing blocks and transactions.	Used for efficient verification of blockchain data.

Key Takeaway

- Cryptography secures blockchain data and transactions.
- Hashing generates unique digital fingerprints for data integrity.
- Merkle Trees organize transaction hashes for efficient verification.
- Together, these technologies make blockchain secure, transparent, and tamper-resistant.

Unit I- Introduction to Blockchain

Blocks, Wallets, and Addresses, Public and Private Key

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Block

A block is a container that stores a group of verified transactions in a blockchain.

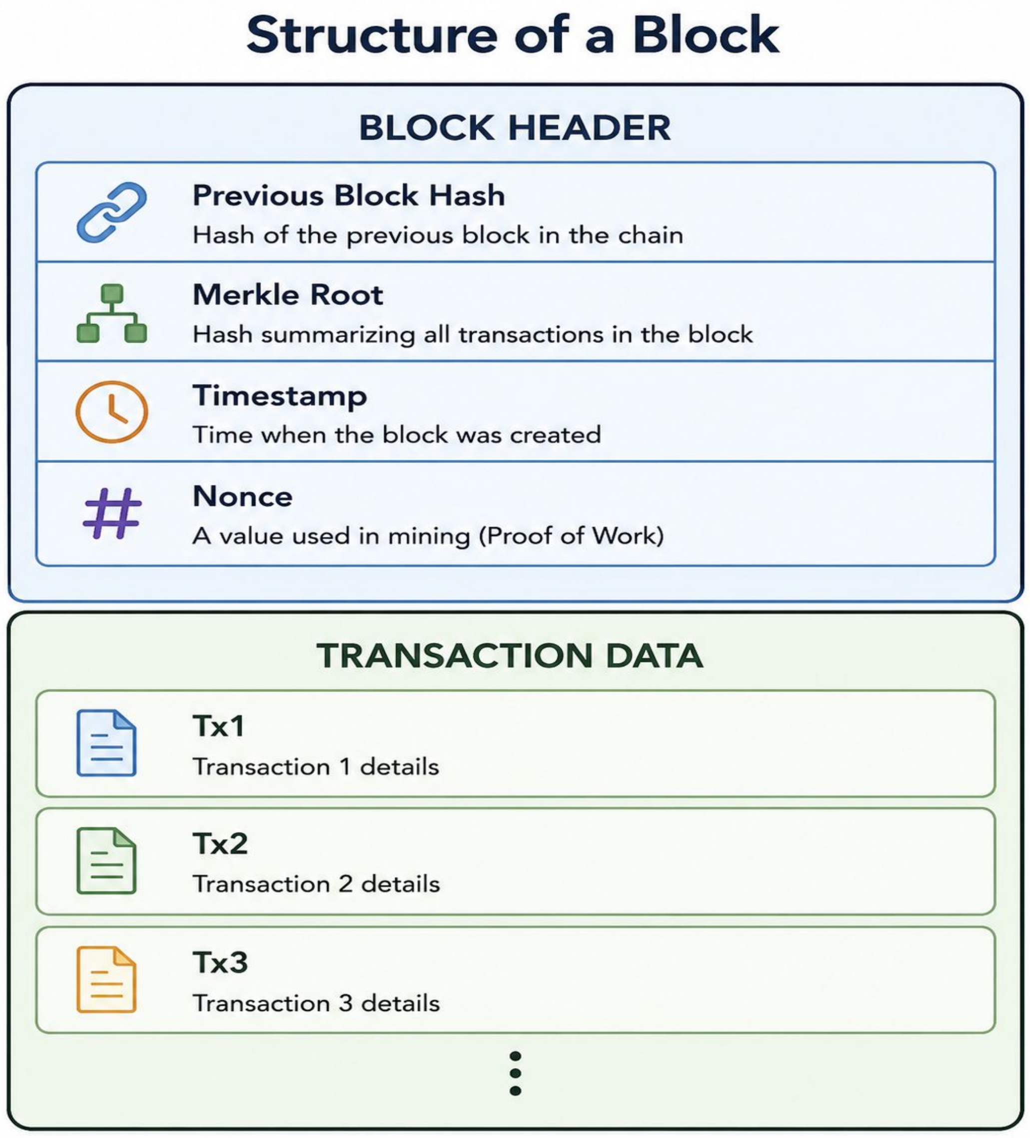
Each block is linked to the previous block using a cryptographic hash, forming a secure chain.

- Stores verified transactions.
- Linked to the previous block.
- Immutable after validation.
- Ensures data integrity.

Key Components of a Block

- Block Header
- Transaction Data
- Previous Block Hash
- Timestamp
- Nonce
- Merkle Root

Structure of a Block



Public Key and Private Key

Blockchain uses asymmetric cryptography, where each user has two keys:

- Private Key – Secret key used to sign transactions.
- Public Key – Shared key used to verify signatures and generate addresses.

Key Points

➢ Keys are mathematically related.
➢ Private key must remain confidential.
➢ Public key can be shared safely.
➢ Used for authentication and secure transactions.

Difference Between Public Key and Private Key

Public Key	Private Key
Shared with others	Kept secret
Used to verify transactions	Used to sign transactions
Can generate wallet addresses	Gives ownership of funds
Safe to distribute	Must never be shared

Wallet

A blockchain wallet is a digital application or device that securely stores a user's cryptographic keys and enables sending, receiving, and managing digital assets.

- Stores public and private keys.
- Sends and receives cryptocurrencies.
- Signs transactions.
- Manages digital assets.

Types of Wallet

- Software Wallet
- Hardware Wallet
- Web Wallet
- Mobile Wallet
- Paper Wallet

Software Wallet

Digital application installed on a computer that stores cryptographic keys and enables users to manage cryptocurrencies.

Key Points

- Installed on desktops or laptops.
- Easy to use and access.
- Suitable for frequent transactions.
- Requires internet access for most operations.

Examples: Exodus, Electrum

Hardware Wallet

Physical device that securely stores private keys offline.

Key Points

- Stores keys offline (cold storage).
- Provides high security against hacking.
- Suitable for long-term storage.
- Requires the device to authorize transactions.

Examples: Ledger Nano X, Trezor

Web Wallet

Online wallet that can be accessed through a web browser without installing software.

Key Points

- Accessible from any internet-connected device.
- No installation required.
- Convenient for quick transactions.
- Security depends on the service provider.

Examples: MetaMask (Browser Extension), Blockchain.com Wallet

Mobile Wallet

Smartphone application that allows users to send, receive, and manage cryptocurrencies.

Key Points

- Installed on smartphones.
- Supports QR code payments.
- Convenient for everyday transactions.
- Often includes biometric authentication.

Examples: Trust Wallet, Coinbase Wallet

Paper Wallet

Printed document containing the public key and private key, usually in the form of text or QR codes.

Key Points

- Completely offline.
- No risk of online hacking.
- Must be stored safely to prevent loss or damage.
- Suitable for long-term storage.

Example: Printed wallet with public and private keys.

Summary on Wallet Types

Wallet Type	Storage	Security	Best For
Software Wallet	Computer	Medium	Regular transactions
Hardware Wallet	Physical Device	Very High	Long-term storage
Web Wallet	Browser/Cloud	Medium	Quick access
Mobile Wallet	Smartphone	High	Daily payments
Paper Wallet	Printed Paper	Very High (Offline)	Backup & long-term storage

Blockchain Address

Unique identifier generated from a public key that is used to receive cryptocurrencies or other digital assets.

Key Points

- Generated from the public key.
- Safe to share.
- Unique for each wallet.
- Used to receive funds.

Example:

0x71C7656EC7ab88b098defB751B7401B5f6d8976F

Key Takeaway

- A **block** stores verified transactions securely.
- **Public and private keys** provide authentication and ownership.
- A **wallet** securely stores cryptographic keys.
- A **wallet address** is used to send and receive digital assets.
- Together, these components enable secure blockchain transactions.

Unit I- Introduction to Blockchain

Introduction to Distributed Ledgers, Introduction to the Decentralized Web

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Distributed Ledger

Shared digital database that is replicated and synchronized across multiple computers (nodes)

where each participant maintains an identical copy of the records.

- Shared among multiple participants.
- No central authority.
- Every node maintains the same ledger.
- Transactions are synchronized across the network.
- Uses cryptography to ensure security and integrity.

Traditional Database vs Distributed Ledger

Traditional Database	Distributed Ledger
Controlled by a central authority	Shared among multiple participants
Single copy of data	Multiple synchronized copies
Easy to modify records	Records are immutable
Lower transparency	High transparency
Single point of failure	No single point of failure

Features of Distributed Ledger

- Decentralized
- Shared Ledger
- Transparency
- Immutability
- Security
- Consensus

How a Distributed Ledger Works

1. A transaction is created.
2. The transaction is broadcast to all participating nodes.
3. Nodes validate the transaction.
4. Consensus is achieved.
5. The transaction is recorded in the distributed ledger.
6. The updated ledger is synchronized across all nodes.

Advantages of Distributed Ledger

- Improved security
- Increased transparency
- High data integrity
- Reduced intermediaries
- Better traceability
- Enhanced reliability

Limitation of Distributed Ledger

- Scalability challenges
- High storage requirements
- Network latency
- Complex governance
- Initial implementation cost

Decentralized Web (Web3)

The next generation of the internet where users own and control their data using blockchain technology instead of relying on centralized organizations.

- User ownership
- Peer-to-peer communication
- Decentralized infrastructure
- Improved privacy
- Transparent interactions

Evolution of the Web

Web 1.0	Web 2.0	Web 3.0
Read	Read & Write	Read, Write & Own
Static websites	Interactive websites	Decentralized applications
Information sharing	User-generated content	User ownership of data

Characteristics of Web3

- Decentralized architecture
- User ownership of data
- Peer-to-peer interactions
- Enhanced privacy
- Smart contract automation
- Token-based economy
- Open and transparent ecosystem

Web2 vs Web3

Web2	Web3
Data owned by companies	Data owned by users
Centralized servers	Distributed blockchain network
Login with username/password	Wallet-based authentication
Platform controls identity	User controls identity
Requires intermediaries	Peer-to-peer interactions

Components of Web3

Core Components

- Blockchain
- Smart Contracts
- Cryptocurrency
- Digital Wallets
- Decentralized Applications (DApps)
- Decentralized Storage
- Digital Identity

Applications of Web3

- Decentralized Finance (DeFi)
- Non-Fungible Tokens (NFTs)
- Blockchain Gaming
- Digital Identity
- Healthcare
- Supply Chain Management
- Education
- Voting Systems
- Metaverse
- Social Media

Future of Web3

Future Trends

- Digital ownership of assets
- AI integrated with blockchain
- Internet of Things (IoT)
- Decentralized finance
- Secure digital identity
- Enterprise blockchain adoption
- Smart cities and e-governance

Case Study

University Digital Certificate System

Scenario:

A university wants to issue digital degree certificates.

Discussion Points

- Should it use a centralized database or a distributed ledger?
- What are the advantages of using blockchain?
- How can students verify certificates securely?

Key Takeaway

- Distributed ledgers securely store and synchronize data across multiple nodes.
- Blockchain is one of the most widely used distributed ledger technologies.
- Web3 enables users to own and control their data.
- Blockchain, smart contracts, and DApps form the foundation of the decentralized web.

THANK YOU

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